

Decidable and enumerable languages

Exercise sheet 2

Exercise 2.1. Is every decidable language enumerable? And vice versa?

Exercise 2.2. Prove that every infinite enumerable language can be enumerated without repetitions.

Exercise 2.3. Which of the following languages are decidable/enumerable?

- (a) The set of all prime numbers (encoded in binary).
- (b) The set of numbers $C = \{x \cdot y \mid x \in A, y \in B\} \subseteq \mathbb{N}$, where $A, B \subseteq \mathbb{N}$ are decidable (respectively, enumerable).
- (c) A language $A \subseteq B$, where $B \subseteq \Sigma^*$ is decidable (respectively, enumerable).

Exercise 2.4. Let A and B be two decidable languages. Which of the following languages are also decidable: $\overline{A}$, $A \cap B$, $A \cup B$? The same question for enumerable A and B .

Exercise 2.5. Which of the following languages are decidable/enumerable?

- (a) HALT-ON-EMPTY—programs that halt on empty input, ε .
- (b) HALT-ON-SOMETHING—programs that halt on some input.
- (c) NEVER-HALT—programs that never halt on any input.

Exercise 2.6. Suppose that for each $i \in \mathbb{N}$, the language $A_i \subseteq \Sigma^*$ is decidable. Is the union $\bigcup_{i \in \mathbb{N}} A_i$ necessarily decidable?

Exercise 2.7. Prove that any two enumerable languages A, B can be decomposed into disjoint enumerable languages. That is, there exist enumerable languages $A', B' \subseteq \Sigma^*$ such that:

$$A' \subseteq A, \quad B' \subseteq B, \quad A' \cup B' = A \cup B, \quad \text{and} \quad A' \cap B' = \emptyset.$$

A language $L \subseteq \Sigma^*$ is called **semidecidable** if there is a program π such that on input $w \in \Sigma^*$:

- if $w \in L$, then π halts and outputs 1;
- if $w \notin L$, then π does not halt.

Exercise 2.8. Prove that a language is enumerable if and only if it is semidecidable.